SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Database Management Systems COURSE CODE: CSA0503
NAME:PURUSHOTHAMAN.M
REG NO:192524517

COURSE OUTCOME COVERED

CO3: *Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity.* (BL2, BL3)

Activity Tool : Case Study (Medium)
Assessment Tool Weightage: 30%

QUESTIONS Total Marks: 50

Q.No	Question	Bloom's Level	Marks
1	Analyze the case: An online shopping platform stores Order(OrderID, CustID, CustName, ProdID, ProdName, Qty, Price). Identify FDs and normalize to 3NF.	BL2 – Understand	**5**
2	A hospital maintains Patient(PID, PName, DocID, DocName, Specialization, WardNo, WardName). Identify anomalies and normalize to BCNF.	BL3 – Apply	**5**
3	Case study: A university stores Enrollment(SID, SName, CourseID, CourseName, Faculty, Grade). Analyze FDs and apply 2NF decomposition.	BL3 – Apply	**5**
4	Given the airline booking case: Booking(FlightNo, PassID, PassName, Seat, DeptCity, ArrCity, Date). Normalize up to BCNF.	BL3 – Apply	**5**
5	Analyze an HR system schema for a multinational company. Identify all functional and transitive dependencies and normalize to 3NF.	BL2 – Understand	**5**

6	Case: Library(MemberID, MemberName, BookID, Title, Author, Genre, BorrowDate). Identify MVDs and decompose to 4NF.	BL3 – Apply	**5**
7	Analyze a telecom billing schema and identify all candidate keys, primary keys, and functional dependencies before normalization.	BL2 – Understand	**5**
8	A retail inventory system has Product(PID, PName, SupplierID, SupplierName, Category, Price, Warehouse). Apply complete normalization.	BL3 – Apply	**5**
9	Study the given poorly normalized schema for a School ERP and propose a fully normalized design up to BCNF with justification.	BL3 – Apply	**5**
10	Given a movie streaming platform schema, identify join dependencies and decompose to 5NF with lossless-join verification.	BL3 – Apply	**5**

Code of Conduct:

I Gudi Rama Krishna (192565028) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:G Rama Krishna

RUBRICS FOR CALCULATION:

Criteria	**Weight a ge**	**Level 4 – Exemplar y**	**Level 3 – Proficien t**	**Level 2 – Developin g**	**Level 1 – Beginnin g**
Understandi ng of Case	25	Thorough understand i ng; clearly identifies all key issues	Good understandi ng with most issues identified	Basic understandi ng; some issues missed	Poor understand i ng; problem not identified correctly
Applicatio n of Concepts	25	Effectively applies relevant concepts to analyze the	Applies appropriat e concepts with minor gaps	Limited application of concepts	Fails to apply relevant concept

		case deeply			s
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

ANSWERS:

1. Online Shopping – 3NF
START
Identify FDs:
OrderID → CustID
CustID → CustName
ProdID → ProdName, Price
(OrderID, ProdID) → Qty

Create Customer(CustID, CustName)
Create Product(ProdID, ProdName, Price)
Create Orders(OrderID, CustID)
Create OrderDetails(OrderID, ProdID, Qty)

```
mysql> SELECT O.OrderID, C.CustName, P.ProdName, OD.Qty, P.Price
    -> FROM Orders O
    -> JOIN Customer C ON O.CustID=C.CustID
    -> JOIN OrderDetails OD ON O.OrderID=OD.OrderID
    -> JOIN Product P ON OD.ProdID=P.ProdID;
+---------+---------+--------+-----+-------+
| OrderID | CustName | ProdName| Qty | Price |
+---------+---------+--------+-----+-------+
|       1 | Ravi     | Laptop  |   1 | 50000 |
|       1 | Ravi     | Mouse   |   2 |   800 |
+---------+---------+--------+-----+-------+
2 rows in set (0.01 sec)
```

END
2. Hospital – BCNF
START
Identify FDs:
PID → PName, DocID, WardNo
DocID → DocName, Specialization
WardNo → WardName

Create Patient(PID, PName, DocID, WardNo)
Create Doctor(DocID, DocName, Specialization)
Create Ward(WardNo, WardName)

```
mysql> SELECT P.PID, P.PName, D.DocName, D.Specialization, W.WardName
    -> FROM Patient P
    -> JOIN Doctor D ON P.DocID=D.DocID
    -> JOIN Ward W ON P.WardNo=W.WardNo;
+-----+-------+---------+----------------+-------------+
| PID | PName | DocName | Specialization | WardName    |
+-----+-------+---------+----------------+-------------+
| 101 | Ravi  | Kumar   | Cardiology     | General     |
| 102 | Sita  | Priya   | Neurology      | ICU         |
+-----+-------+---------+----------------+-------------+
2 rows in set (0.01 sec)
```

END

3. University Enrollment – 2NF

START

Identify FDs:

SID → SName

CourseID → CourseName, Faculty

(SID, CourseID) → Grade

Create Student(SID, SName)

Create Course(CourseID, CourseName, Faculty)

Create Enrollment(SID, CourseID, Grade)

END

```
mysql> SELECT E.SID, S.SName, C.CID, C.CName, C.Faculty, E.Grade
    -> FROM Enrollment E
    -> JOIN Student S ON E.SID=S.SID
    -> JOIN Course C ON E.CID=C.CID;
+-----+-------+-----+-------------------+------------+-------+
| SID | SName | CID | CName             | Faculty    | Grade |
+-----+-------+-----+-------------------+------------+-------+
| 101 | Arun  | 201 | DBMS              | Kumar      | A     |
| 101 | Arun  | 202 | Operating Systems | Ravi       | B     |
| 102 | Priya | 201 | DBMS              | Kumar      | A     |
| 102 | Priya | 202 | Operating Systems | Ravi       | A     |
+-----+-------+-----+-------------------+------------+-------+
4 rows in set (0.01 sec)
```

4. Airline Booking – BCNF

START

Identify FDs:

FlightNo → DeptCity, ArrCity

PassID → PassName

(FlightNo, Date, Seat) → PassID

Create Flight(FlightNo, DeptCity, ArrCity)

Create Passenger(PassID, PassName)

Create Booking(FlightNo, PassID, Seat, Date)

END

```
mysql> SELECT B.FlightNo, B.PassID, P.PassName, B.Seat,
    -> F.DeptCity, F.ArrCity, B.BookingDate
    -> FROM Booking B
    -> JOIN Passenger P ON B.PassID=P.PassID
    -> JOIN Flight F ON B.FlightNo=F.FlightNo;
+----------+--------+----------+------+----------+---------+-------------+
| FlightNo | PassID | PassName | Seat | DeptCity | ArrCity | BookingDate |
+----------+--------+----------+------+----------+---------+-------------+
| AI101    |      1 | Ravi     | 12A  | Chennai  | Delhi   | 2026-08-20  |
| AI102    |      2 | Kiran    | 10B  | Mumbai   | Delhi   | 2026-08-21  |
+----------+--------+----------+------+----------+---------+-------------+
2 rows in set (0.01 sec)
```

5. HR System – 3NF
START
Identify FDs:
EmpID → EmpName, DeptID, JobID
DeptID → DeptName, Location
JobID → JobTitle, SalaryGrade

Create Employee(EmpID, EmpName, DeptID, JobID)
Create Department(DeptID, DeptName, Location)
Create Job(JobID, JobTitle, SalaryGrade)

END

```
mysql> SELECT E.EmpID, E.EmpName, D.DeptName, D.Location,
    -> J.JobTitle, J.SalaryGrade
    -> FROM Employee E
    -> JOIN Department D ON E.DeptID=D.DeptID
    -> JOIN Job J ON E.JobID=J.JobID;
+-------+---------+----------+-----------+-------------------+------------+
| EmpID | EmpName | DeptName | Location  | JobTitle          | SalaryGrade|
+-------+---------+----------+-----------+-------------------+------------+
| 1001  | Ravi    | IT       | Chennai   | Software Engineer | G5         |
| 1002  | Sita    | HR       | Bangalore | HR Manager        | G7         |
+-------+---------+----------+-----------+-------------------+------------+
2 rows in set (0.01 sec)
```

6. Library – 4NF
START
Identify MVDs:
BookID →→ Author
BookID →→ Genre

Create Book(BookID, Title)
Create BookAuthor(BookID, Author)
Create BookGenre(BookID, Genre)
Create Member(MemberID, MemberName)

Create Borrow(MemberID, BookID, BorrowDate)

END

```
mysql> SELECT B.BookID, B.Title, A.Author, G.Genre
    -> FROM Book B
    -> JOIN BookAuthor A ON B.BookID=A.BookID
    -> JOIN BookGenre G ON B.BookID=G.BookID;
+--------+------------------+--------------+------------+
| BookID | Title            | Author       | Genre      |
+--------+------------------+--------------+------------+
| 101    | Database Systems   | Korth          | Database    |
| 101    | Database Systems   | Korth          | Education   |
| 101    | Database Systems   | Ramakrishnan   | Database    |
| 101    | Database Systems   | Ramakrishnan   | Education   |
+--------+------------------+--------------+------------+
4 rows in set (0.01 sec)
```

7. Telecom Billing
START
Identify candidate keys:
CustomerID
Phone
BillID

Identify FDs:
CustomerID → CustomerName, Phone
Phone → CustomerID, CustomerName
BillID → CustomerID, Amount

Create Customer(CustomerID, CustomerName, Phone)
Create Bill(BillID, CustomerID, Amount)

END

```
mysql> SELECT B.BillID, C.CustomerID, C.CustomerName,
    -> C.Phone, B.Amount
    -> FROM Bill B JOIN Customer C ON B.CustomerID=C.CustomerID;
+--------+------------+--------------+------------+--------+
| BillID | CustomerID | CustomerName | Phone      | Amount |
+--------+------------+--------------+------------+--------+
|   1001 |        101 | Ravi         | 9876543210 |    599 |
|   1002 |        102 | Kiran        | 9876543211 |    299 |
+--------+------------+--------------+------------+--------+
2 rows in set (0.01 sec)
```

8. Retail Inventory – Normalization
START
Identify FDs:
PID → PName, SupplierID, Category, Price
SupplierID → SupplierName

Create Product(PID, PName, SupplierID, Category, Price)
Create Supplier(SupplierID, SupplierName)
Create Warehouse(WarehouseID, WarehouseName)

END

```
mysql> SELECT P.PID, P.PName, S.SupplierName,
    -> P.Category, P.Price, P.Warehouse
    -> FROM Product P JOIN Supplier S ON P.SupplierID=S.SupplierID;
+-----+--------+--------------+-------------+-------+-----------+
| PID | PName  | SupplierName | Category    | Price | Warehouse |
+-----+--------+--------------+-------------+-------+-----------+
| 501 | Laptop | ABC          | Electronics | 50000 | Chennai   |
| 502 | Mouse  | XYZ          | Accessories |   800 | Chennai   |
+-----+--------+--------------+-------------+-------+-----------+
2 rows in set (0.01 sec)
```

9. School ERP – BCNF
START
Identify FDs:
StudentID → StudentName, DeptID
DeptID → DeptName
SubjectID → SubjectName, TeacherID
TeacherID → TeacherName

Create Student(StudentID, StudentName, DeptID)
Create Department(DeptID, DeptName)
Create Subject(SubjectID, SubjectName, TeacherID)
Create Teacher(TeacherID, TeacherName)
Create StudentSubject(StudentID, SubjectID, Marks)

END

```
mysql> SELECT S.StudentID, S.StudentName, D.DeptName,
    -> SU.SubjectName, T.TeacherName, SS.Marks
    -> FROM StudentSubject SS
    -> JOIN Student S ON SS.StudentID=S.StudentID
    -> JOIN Department D ON S.DeptID=D.DeptID
    -> JOIN Subject SU ON SS.SubjectID=SU.SubjectID
    -> JOIN Teacher T ON SU.TeacherID=T.TeacherID;
+-----------+-------------+----------------+-------------------+-------------+-------+
| StudentID | StudentName | DeptName       | SubjectName       | TeacherName | Marks |
+-----------+-------------+----------------+-------------------+-------------+-------+
| 1         | Ravi        | Computer Science| DBMS              | Kumar       | 85    |
| 1         | Ravi        | Computer Science| Operating Systems | Priya       | 90    |
| 2         | Kiran       | Computer Science| DBMS              | Kumar       | 78    |
+-----------+-------------+----------------+-------------------+-------------+-------+
3 rows in set (0.01 sec)
```

10. Movie Streaming – 5NF
START
Identify Join Dependency:
Movie, Actor, Platform

Create Movie(MovieID, MovieName)
Create Actor(ActorID, ActorName)
Create Platform(PlatformID, PlatformName)
Create MovieActor(MovieID, ActorID)
Create MoviePlatform(MovieID, PlatformID)
Create ActorPlatform(ActorID, PlatformID)

Verify lossless join

END

```
mysql> SELECT M.MovieName, A.ActorName, P.PlatformName
    -> FROM Movie M
    -> JOIN MovieActor MA ON M.MovieID=MA.MovieID
    -> JOIN Actor A ON MA.ActorID=A.ActorID
    -> JOIN MoviePlatform MP ON M.MovieID=MP.MovieID
    -> JOIN Platform P ON MP.PlatformID=P.PlatformID;
+-----------+-----------+--------------+
| MovieName | ActorName  | PlatformName |
+-----------+-----------+--------------+
| RRR       | NTR        | Netflix      |
| RRR       | Ram Charan | Netflix      |
| RRR       | NTR        | Prime Video  |
+-----------+-----------+--------------+
3 rows in set (0.01 sec)
```